

V3 empirical study · Instruction

Six models · three underlyings · twelve expiry windows · 1-day hourly calibration

What this report is

- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of six models tracks realized stock price S_t most closely on the evaluation window, and does the ranking change across companies and the twelve windows?
- This report does not price options. Option RMSE lives in the companion decision PDF.
- In every table the BEST row is the lowest percentage RMSE of the p50 path versus realized S_t . That row is bold and shaded green; Mark ranks 1–6.

Experimental design (held fixed for every model)

- Models: GBM, GARCH(1,1), Heston (no jumps), Merton jump-diffusion, GARCH–Merton, Heston–Merton (Bates).
- Companies: SPY, AAPL, MSFT. Each name is calibrated and simulated separately.
- Windows: twelve Friday-before-expiry evaluation windows (Oct 2022–Sep 2023). This is the 1-day study.
- Calibration: 1-day of 1-minute RTH returns (and listed quotes for Heston) ending at each hourly stamp 09:59–15:59. No look-ahead.
- P-measure: keep the lookback drift μ . Do not replace $\mu \rightarrow r$ (that replacement is only for LSM option pricing).

V3 empirical study · Instruction (continued)

Six models · three underlyings · twelve expiry windows · 1-day hourly calibration

Path construction

- One rolling path cloud per ticker $\times$ window $\times$ model, $n_{\text{paths}} = 10000$, seed 42.
- Clock: first RTH bar of the window through Friday 15:59 in 1-minute steps. Not to listed option expiry.
- Start at the observed open S_0 . Refresh parameters at each hourly stamp with data $\leq$ that stamp.
- At each clock time, summarize the cloud by percentiles p_{10} , p_{25} , p_{50} , p_{75} , p_{90} . The reported path is p_{50} , not the Monte Carlo mean.

Metrics

- Primary: $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((p_{50} - S_t)/S_t)^2)}$. Ranking uses this only (lowest is Mark 1).
- $\text{MAE} = \text{mean } |p_{50} - S_t|$ (dollars). $\text{Bias \$} = \text{mean}(p_{50} - S_t)$. $\text{Bias\%} = 100 \times \text{mean}((p_{50} - S_t)/S_t)$.
- $\text{ICP } 25\text{--}75 = \text{share of historical } S_t \text{ inside } [p_{25}, p_{75}]$. $\text{ABW} = \text{mean}(p_{75} - p_{25})$. $\text{ICP } 10\text{--}90 = \text{share inside } [p_{10}, p_{90}]$ (method/appendix; not a table column).
- ICP is higher-better and is not used for the green row.

Page 3 · Table block · SPY · six models × four expiry windows

P-measure rolling cloud, reported path = p50. Bold green row = lowest RMSE% vs realized S_t . 1-day lookback, hourly rolling, $\Delta t = 1$ min, $n_{\text{paths}}=10000$, $\text{seed}=42$.

Table SPY-1 SPY · Oct 2022 listed expiry 2022-10-21
2022-10-14 09:30 → 2022-10-14 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	2.91	10.2043	+10.1916	+2.79	2.3%	7.3404	2
GARCH	3.03	10.5467	+10.5318	+2.88	2.1%	6.7293	6
Heston	2.97	10.4122	+10.4002	+2.84	1.8%	6.2224	4
Merton	2.67	9.3581	+9.3424	+2.55	1.8%	7.1763	1
GARCH-Merton	3.01	10.4451	+10.4285	+2.85	2.1%	7.6177	5
Heston-Merton	2.93	10.2758	+10.2626	+2.81	1.5%	5.0359	3

Table SPY-2 SPY · Nov 2022 listed expiry 2022-11-18
2022-11-11 09:30 → 2022-11-11 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.33	1.0422	+0.8966	+0.24	81.0%	3.5399	1
GARCH	0.47	1.5132	+1.3930	+0.37	79.2%	4.1913	6
Heston	0.42	1.3570	+1.2446	+0.33	86.7%	4.1509	3
Merton	0.44	1.4487	+1.3532	+0.36	63.8%	3.4973	4
GARCH-Merton	0.46	1.4991	+1.3845	+0.37	82.8%	4.3513	5
Heston-Merton	0.41	1.3137	+1.2030	+0.32	88.5%	4.2742	2

Table SPY-3 SPY · Dec 2022 listed expiry 2022-12-16
2022-12-09 09:30 → 2022-12-09 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.26	0.7816	-0.1367	-0.04	79.0%	3.0792	3
GARCH	0.26	0.7918	-0.6349	-0.17	76.9%	3.1764	1
Heston	0.31	0.9367	+0.2426	+0.06	98.2%	5.2194	6
Merton	0.28	0.8333	+0.0899	+0.02	83.6%	2.9956	4
GARCH-Merton	0.26	0.8032	-0.6506	-0.17	77.4%	3.2193	2
Heston-Merton	0.30	0.9083	+0.1483	+0.04	98.2%	5.3118	5

Table SPY-4 SPY · Jan 2023 listed expiry 2023-01-20
2023-01-13 09:30 → 2023-01-13 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.29	0.9715	-0.9521	-0.26	84.6%	3.2451	6
GARCH	0.21	0.6453	-0.6028	-0.16	84.1%	3.2944	2
Heston	0.27	0.9060	-0.8857	-0.24	72.3%	2.3326	3
Merton	0.28	0.8965	-0.8795	-0.24	80.3%	3.1809	5
GARCH-Merton	0.21	0.6444	-0.5967	-0.16	86.4%	3.3130	1
Heston-Merton	0.28	0.9292	-0.9104	-0.25	49.0%	1.8486	4

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE and Bias \$ are in stock-price dollars. ICP 25-75 is the share of historical S_t inside $[p25, p75]$. ABW = $\text{mean}(p75 - p25)$. Ranking uses RMSE% only.

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P-measure rolling cloud, reported path = p50. Bold green row = lowest RMSE% vs realized S_t. 1-day lookback, hourly rolling, Δt = 1 min, n_paths=10000, seed=42.

Table SPY-5 SPY · Feb 2023 listed expiry 2023-02-17
2023-02-10 09:30 → 2023-02-10 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.76	2.6259	-2.5907	-0.67	11.5%	3.2267	5
GARCH	0.63	2.0778	-2.0217	-0.52	24.9%	2.9262	1
Heston	0.71	2.4678	-2.4288	-0.63	35.1%	4.5933	4
Merton	0.92	3.1643	-3.1327	-0.81	8.2%	3.1598	6
GARCH-Merton	0.64	2.0951	-2.0407	-0.53	29.2%	3.1693	2
Heston-Merton	0.70	2.4302	-2.3931	-0.62	33.6%	4.5534	3

Table SPY-6 SPY · Mar 2023 listed expiry 2023-03-17
2023-03-10 09:30 → 2023-03-10 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.71	2.3231	-2.1660	-0.56	40.8%	4.0319	3
GARCH	0.87	2.8312	-2.6434	-0.69	26.4%	3.7092	4
Heston	0.60	1.9300	-1.7378	-0.45	65.1%	5.3339	1
Merton	1.02	3.3937	-3.2313	-0.84	22.8%	3.9245	6
GARCH-Merton	0.88	2.8844	-2.6929	-0.70	29.2%	4.0515	5
Heston-Merton	0.64	2.0743	-1.9015	-0.50	63.6%	5.4469	2

Table SPY-7 SPY · Apr 2023 listed expiry 2023-04-21
2023-04-14 09:30 → 2023-04-14 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.77	2.8724	+2.7040	+0.67	2.1%	2.0633	3
GARCH	0.63	2.3199	+2.1245	+0.53	3.3%	1.2533	1
Heston	0.83	3.1147	+2.9532	+0.74	4.4%	2.8279	5
Merton	0.97	3.5897	+3.4249	+0.85	2.1%	2.0421	6
GARCH-Merton	0.63	2.3264	+2.1324	+0.53	3.3%	1.5383	2
Heston-Merton	0.81	3.0310	+2.8674	+0.72	7.2%	2.9560	4

Table SPY-8 SPY · May 2023 listed expiry 2023-05-19
2023-05-12 09:30 → 2023-05-12 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.60	2.1928	+2.1889	+0.55	13.8%	2.0454	1
GARCH	0.65	2.3829	+2.3792	+0.60	12.6%	2.0817	5
Heston	0.62	2.2532	+2.2499	+0.56	17.9%	2.7619	3
Merton	0.77	2.8445	+2.8412	+0.71	6.2%	2.0339	6
GARCH-Merton	0.65	2.3821	+2.3789	+0.60	13.1%	2.2489	4
Heston-Merton	0.61	2.2198	+2.2164	+0.56	17.4%	2.7224	2

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE and Bias \$ are in stock-price dollars. ICP 25-75 is the share of historical S_t inside [p25, p75]. ABW = $\text{mean}(p75 - p25)$. Ranking uses RMSE% only.

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P-measure rolling cloud, reported path = p50. Bold green row = lowest RMSE% vs realized S_t . 1-day lookback, hourly rolling, $\Delta t = 1$ min, $n_{\text{paths}}=10000$, $\text{seed}=42$.

Table SPY-9 SPY · Jun 2023 listed expiry 2023-06-16
2023-06-09 09:30 → 2023-06-09 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.30	1.1360	+0.7284	+0.18	27.4%	1.7330	1
GARCH	0.35	1.3040	+0.8234	+0.20	4.4%	1.4341	4
Heston	0.32	1.2171	+0.8286	+0.20	54.4%	2.4075	3
Merton	0.38	1.4511	+0.9774	+0.24	5.6%	1.7177	6
GARCH-Merton	0.35	1.3230	+0.8402	+0.20	8.7%	1.5882	5
Heston-Merton	0.31	1.1941	+0.7951	+0.19	56.4%	2.3942	2

Table SPY-10 SPY · Jul 2023 listed expiry 2023-07-21
2023-07-14 09:30 → 2023-07-14 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.29	1.1055	+1.0034	+0.23	16.9%	1.4891	3
GARCH	0.26	0.9770	+0.8664	+0.20	9.0%	1.0375	2
Heston	0.34	1.2846	+1.1938	+0.28	49.2%	2.2301	5
Merton	0.31	1.1920	+1.0975	+0.25	13.8%	1.4765	4
GARCH-Merton	0.26	0.9739	+0.8650	+0.20	15.4%	1.2658	1
Heston-Merton	0.34	1.2979	+1.2073	+0.28	44.1%	2.1639	6

Table SPY-11 SPY · Aug 2023 listed expiry 2023-08-18
2023-08-11 09:30 → 2023-08-11 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.58	2.3647	-2.3625	-0.56	6.4%	3.2063	3
GARCH	0.74	2.9329	-2.9306	-0.69	2.8%	2.2537	5
Heston	0.56	2.2920	-2.2898	-0.54	6.9%	3.1223	1
Merton	0.60	2.4287	-2.4265	-0.57	5.9%	3.1835	4
GARCH-Merton	0.75	2.9359	-2.9334	-0.69	3.6%	2.5419	6
Heston-Merton	0.57	2.3118	-2.3096	-0.54	5.6%	3.1024	2

Table SPY-12 SPY · Sep 2023 listed expiry 2023-09-15
2023-09-08 09:30 → 2023-09-08 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.21	0.7513	+0.1980	+0.05	63.6%	1.8953	3
GARCH	0.18	0.6507	-0.4882	-0.12	52.6%	1.3610	1
Heston	0.24	0.8336	+0.3456	+0.08	65.4%	2.0240	5
Merton	0.25	0.8698	+0.3445	+0.08	51.3%	1.8571	6
GARCH-Merton	0.18	0.6568	-0.4968	-0.12	55.6%	1.5650	2
Heston-Merton	0.23	0.8122	+0.3133	+0.07	62.8%	1.9382	4

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE and Bias \$ are in stock-price dollars. ICP 25-75 is the share of historical S_t inside $[p25, p75]$. ABW = $\text{mean}(p75 - p25)$. Ranking uses RMSE% only.

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P-measure rolling cloud, reported path = p50. Bold green row = lowest RMSE% vs realized S_t . 1-day lookback, hourly rolling, $\Delta t = 1$ min, $n_{\text{paths}}=10000$, $\text{seed}=42$.

Table AAPL-1 AAPL · Oct 2022 listed expiry 2022-10-21
2022-10-14 09:30 → 2022-10-14 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	3.86	5.4210	+5.4210	+3.76	0.3%	3.6204	2
GARCH	3.99	5.5714	+5.5714	+3.87	0.3%	3.3699	6
Heston	3.89	5.4669	+5.4669	+3.79	0.3%	3.6744	3
Merton	3.04	4.2932	+4.2932	+2.98	0.3%	3.5568	1
GARCH-Merton	3.95	5.5195	+5.5195	+3.83	0.3%	3.9592	5
Heston-Merton	3.92	5.4973	+5.4973	+3.81	0.3%	3.4971	4

Table AAPL-2 AAPL · Nov 2022 listed expiry 2022-11-18
2022-11-11 09:30 → 2022-11-11 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.62	0.6408	-0.0589	-0.04	80.3%	2.4240	3
GARCH	0.56	0.5747	+0.0374	+0.03	80.8%	2.2077	2
Heston	0.63	0.6477	+0.0294	+0.02	83.1%	2.9594	5
Merton	0.64	0.6437	+0.0790	+0.06	75.1%	2.4125	6
GARCH-Merton	0.55	0.5606	+0.0235	+0.02	83.1%	2.5432	1
Heston-Merton	0.62	0.6322	-0.0052	-0.00	83.1%	2.9089	4

Table AAPL-3 AAPL · Dec 2022 listed expiry 2022-12-16
2022-12-09 09:30 → 2022-12-09 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	1.03	1.2183	-1.0291	-0.71	49.5%	2.0853	3
GARCH	1.32	1.6895	-1.6330	-1.14	19.5%	1.8993	5
Heston	0.99	1.1576	-0.9413	-0.65	67.9%	3.3977	1
Merton	1.05	1.2744	-1.1262	-0.78	43.3%	2.0472	4
GARCH-Merton	1.33	1.6965	-1.6394	-1.14	22.6%	2.1272	6
Heston-Merton	1.01	1.1883	-0.9883	-0.69	66.9%	3.3798	2

Table AAPL-4 AAPL · Jan 2023 listed expiry 2023-01-20
2023-01-13 09:30 → 2023-01-13 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.76	0.9262	-0.9262	-0.71	68.7%	1.9347	5
GARCH	0.72	0.8687	-0.8687	-0.67	71.0%	2.0243	2
Heston	0.73	0.8842	-0.8842	-0.68	71.8%	2.2486	3
Merton	1.01	1.2525	-1.2525	-0.97	23.8%	1.9034	6
GARCH-Merton	0.72	0.8667	-0.8667	-0.67	72.1%	2.1004	1
Heston-Merton	0.75	0.9243	-0.9243	-0.71	71.5%	2.1957	4

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE and Bias \$ are in stock-price dollars. ICP 25-75 is the share of historical S_t inside $[p25, p75]$. ABW = $\text{mean}(p75 - p25)$. Ranking uses RMSE% only.

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P-measure rolling cloud, reported path = p50. Bold green row = lowest RMSE% vs realized S_t. 1-day lookback, hourly rolling, Δt = 1 min, n_paths=10000, seed=42.

Table AAPL-5 AAPL · Feb 2023 listed expiry 2023-02-17
2023-02-10 09:30 → 2023-02-10 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.96	1.2524	-1.2294	-0.83	28.2%	1.9909	3
GARCH	1.42	1.8219	-1.8022	-1.22	9.5%	1.7603	5
Heston	0.90	1.1611	-1.1355	-0.77	86.4%	3.1447	1
Merton	1.09	1.3713	-1.3441	-0.91	30.0%	1.9446	4
GARCH-Merton	1.43	1.8286	-1.8080	-1.22	11.0%	1.8678	6
Heston-Merton	0.93	1.1997	-1.1757	-0.79	84.1%	3.1483	2

Table AAPL-6 AAPL · Mar 2023 listed expiry 2023-03-17
2023-03-10 09:30 → 2023-03-10 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.64	0.8538	-0.7549	-0.50	63.6%	1.9648	3
GARCH	0.82	1.0892	-0.9823	-0.66	50.3%	2.2102	6
Heston	0.60	0.7905	-0.6856	-0.46	90.3%	3.1499	2
Merton	0.76	1.0024	-0.8954	-0.60	45.6%	1.9363	4
GARCH-Merton	0.82	1.0766	-0.9675	-0.65	52.3%	2.2639	5
Heston-Merton	0.60	0.7813	-0.6784	-0.45	90.8%	3.1454	1

Table AAPL-7 AAPL · Apr 2023 listed expiry 2023-04-21
2023-04-14 09:30 → 2023-04-14 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.98	1.4754	+1.3455	+0.82	3.8%	1.4144	2
GARCH	1.09	1.6156	+1.4815	+0.90	3.1%	1.1214	5
Heston	1.01	1.5144	+1.3876	+0.85	4.1%	1.4398	4
Merton	0.98	1.4646	+1.3284	+0.81	5.1%	1.3947	1
GARCH-Merton	1.09	1.6238	+1.4900	+0.91	3.1%	1.2300	6
Heston-Merton	1.01	1.5096	+1.3798	+0.84	3.8%	1.4241	3

Table AAPL-8 AAPL · May 2023 listed expiry 2023-05-19
2023-05-12 09:30 → 2023-05-12 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.87	1.3490	+1.3437	+0.80	8.5%	1.2248	1
GARCH	0.89	1.3701	+1.3648	+0.82	8.7%	1.3824	3
Heston	0.91	1.4133	+1.4081	+0.84	13.3%	1.7545	5
Merton	0.92	1.4197	+1.4143	+0.85	6.2%	1.2102	6
GARCH-Merton	0.89	1.3781	+1.3728	+0.82	8.7%	1.3948	4
Heston-Merton	0.88	1.3630	+1.3578	+0.81	14.9%	1.7503	2

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE and Bias \$ are in stock-price dollars. ICP 25-75 is the share of historical S_t inside [p25, p75]. ABW = $\text{mean}(p75 - p25)$. Ranking uses RMSE% only.

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P-measure rolling cloud, reported path = p50. Bold green row = lowest RMSE% vs realized S_t. 1-day lookback, hourly rolling, Δt = 1 min, n_paths=10000, seed=42.

Table AAPL-9 AAPL · Jun 2023 listed expiry 2023-06-16
2023-06-09 09:30 → 2023-06-09 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.54	0.8220	+0.8115	+0.48	43.6%	1.4902	1
GARCH	0.60	0.9122	+0.9016	+0.53	29.0%	1.3195	2
Heston	0.63	0.9628	+0.9533	+0.56	96.4%	2.9430	4
Merton	0.83	1.2698	+1.2612	+0.74	11.3%	1.4670	6
GARCH-Merton	0.61	0.9247	+0.9141	+0.54	40.0%	1.5093	3
Heston-Merton	0.64	0.9714	+0.9613	+0.56	96.2%	2.9487	5

Table AAPL-10 AAPL · Jul 2023 listed expiry 2023-07-21
2023-07-14 09:30 → 2023-07-14 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.23	0.3365	+0.2431	+0.14	77.4%	1.1344	3
GARCH	0.21	0.3125	+0.1733	+0.10	75.6%	0.9686	2
Heston	0.24	0.3538	+0.2650	+0.15	96.7%	1.7334	5
Merton	0.27	0.4093	+0.3390	+0.19	69.7%	1.1158	6
GARCH-Merton	0.20	0.3074	+0.1678	+0.09	81.3%	1.1403	1
Heston-Merton	0.23	0.3523	+0.2653	+0.15	96.4%	1.7146	4

Table AAPL-11 AAPL · Aug 2023 listed expiry 2023-08-18
2023-08-11 09:30 → 2023-08-11 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.66	1.0302	-1.0105	-0.61	30.3%	1.6681	4
GARCH	0.93	1.3948	-1.3746	-0.83	17.7%	1.9515	5
Heston	0.56	0.8709	-0.8487	-0.51	78.5%	2.2288	1
Merton	0.62	0.9516	-0.9297	-0.56	39.5%	1.6416	3
GARCH-Merton	0.94	1.4043	-1.3835	-0.84	19.5%	2.0041	6
Heston-Merton	0.59	0.9192	-0.8981	-0.54	71.8%	2.1743	2

Table AAPL-12 AAPL · Sep 2023 listed expiry 2023-09-15
2023-09-08 09:30 → 2023-09-08 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.53	0.6147	+0.4188	+0.25	74.1%	1.9111	3
GARCH	0.43	0.5294	+0.1886	+0.11	92.1%	2.2009	1
Heston	0.60	0.7061	+0.5494	+0.33	95.9%	3.0951	5
Merton	0.60	0.6932	+0.4728	+0.28	72.3%	1.8704	6
GARCH-Merton	0.44	0.5372	+0.1924	+0.12	93.3%	2.3010	2
Heston-Merton	0.58	0.6823	+0.5049	+0.30	93.1%	2.8912	4

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE and Bias \$ are in stock-price dollars. ICP 25-75 is the share of historical S_t inside [p25, p75]. ABW = $\text{mean}(p75 - p25)$. Ranking uses RMSE% only.

Page 9 · Table block · MSFT · six models × four expiry windows

P-measure rolling cloud, reported path = p50. Bold green row = lowest RMSE% vs realized S_t. 1-day lookback, hourly rolling, Δt = 1 min, n_paths=10000, seed=42.

Table MSFT-1 MSFT · Oct 2022 listed expiry 2022-10-21
2022-10-14 09:30 → 2022-10-14 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	3.25	7.5375	+7.5375	+3.14	1.0%	6.0933	5
GARCH	2.92	6.7949	+6.7949	+2.83	0.5%	4.9842	3
Heston	3.27	7.5908	+7.5908	+3.16	0.8%	5.8050	6
Merton	2.71	6.3218	+6.3218	+2.63	1.0%	5.9220	1
GARCH-Merton	2.87	6.6894	+6.6894	+2.78	0.8%	5.9678	2
Heston-Merton	3.23	7.5132	+7.5132	+3.13	0.8%	5.6913	4

Table MSFT-2 MSFT · Nov 2022 listed expiry 2022-11-18
2022-11-11 09:30 → 2022-11-11 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.50	1.0054	+0.2474	+0.10	76.4%	3.4730	3
GARCH	0.50	0.9927	+0.3538	+0.15	75.4%	3.1281	4
Heston	0.54	1.0586	+0.4476	+0.19	87.9%	4.5819	6
Merton	0.45	0.9019	+0.1103	+0.05	78.5%	3.4260	1
GARCH-Merton	0.49	0.9729	+0.3299	+0.14	79.7%	3.3493	2
Heston-Merton	0.52	1.0370	+0.3605	+0.15	87.2%	4.5155	5

Table MSFT-3 MSFT · Dec 2022 listed expiry 2022-12-16
2022-12-09 09:30 → 2022-12-09 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.27	0.5271	+0.0771	+0.03	85.6%	3.5495	1
GARCH	0.29	0.6345	-0.2836	-0.11	83.8%	2.8667	5
Heston	0.29	0.5899	+0.2597	+0.10	95.4%	5.9180	4
Merton	0.36	0.7428	+0.5814	+0.23	83.8%	3.4393	6
GARCH-Merton	0.29	0.6058	-0.2269	-0.09	84.4%	3.1671	3
Heston-Merton	0.28	0.5546	+0.1773	+0.07	95.4%	5.9296	2

Table MSFT-4 MSFT · Jan 2023 listed expiry 2023-01-20
2023-01-13 09:30 → 2023-01-13 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.46	0.9498	+0.1918	+0.08	85.6%	3.4800	4
GARCH	0.52	0.9928	-0.4789	-0.19	77.4%	2.8070	6
Heston	0.46	0.9504	+0.2165	+0.09	88.5%	3.9687	3
Merton	0.44	0.8644	+0.4387	+0.18	84.4%	3.4538	1
GARCH-Merton	0.52	0.9900	-0.4633	-0.19	86.4%	3.1748	5
Heston-Merton	0.45	0.9321	+0.2249	+0.09	87.4%	3.8119	2

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE and Bias \$ are in stock-price dollars. ICP 25-75 is the share of historical S_t inside [p25, p75]. ABW = $\text{mean}(p75 - p25)$. Ranking uses RMSE% only.

Page 10 · Table block · MSFT · six models × four expiry windows

P-measure rolling cloud, reported path = p50. Bold green row = lowest RMSE% vs realized S_t. 1-day lookback, hourly rolling, Δt = 1 min, n_paths=10000, seed=42.

Table MSFT-5 MSFT · Feb 2023 listed expiry 2023-02-17
2023-02-10 09:30 → 2023-02-10 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	1.15	2.5741	-2.5126	-0.98	22.8%	3.6739	4
GARCH	1.35	2.9558	-2.8916	-1.13	19.5%	3.6639	5
Heston	1.10	2.4715	-2.4053	-0.94	59.7%	5.5639	2
Merton	1.00	2.2110	-2.1361	-0.83	37.2%	3.6323	1
GARCH-Merton	1.36	2.9752	-2.9105	-1.13	20.3%	3.6708	6
Heston-Merton	1.14	2.5494	-2.4864	-0.97	54.6%	5.5223	3

Table MSFT-6 MSFT · Mar 2023 listed expiry 2023-03-17
2023-03-10 09:30 → 2023-03-10 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.95	2.1824	-1.9311	-0.78	33.1%	3.4597	3
GARCH	1.34	3.0454	-2.8087	-1.13	7.4%	3.1410	5
Heston	0.88	2.0197	-1.7582	-0.71	82.3%	5.6646	1
Merton	1.23	2.7833	-2.5181	-1.01	10.8%	3.4129	4
GARCH-Merton	1.34	3.0496	-2.8076	-1.13	7.9%	3.4813	6
Heston-Merton	0.92	2.1067	-1.8515	-0.74	79.0%	5.6051	2

Table MSFT-7 MSFT · Apr 2023 listed expiry 2023-04-21
2023-04-14 09:30 → 2023-04-14 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.98	2.5007	+2.4003	+0.85	3.1%	2.3823	4
GARCH	0.79	1.9659	+1.8352	+0.65	7.4%	1.6964	2
Heston	0.98	2.5054	+2.4070	+0.85	9.2%	2.9759	5
Merton	0.96	2.4249	+2.3065	+0.82	6.7%	2.3828	3
GARCH-Merton	0.79	1.9586	+1.8243	+0.65	9.0%	2.0134	1
Heston-Merton	0.98	2.5144	+2.4121	+0.86	10.3%	3.0317	6

Table MSFT-8 MSFT · May 2023 listed expiry 2023-05-19
2023-05-12 09:30 → 2023-05-12 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.61	1.6702	+1.6568	+0.57	22.3%	2.4588	1
GARCH	0.67	1.8656	+1.8656	+0.64	15.1%	2.3504	3
Heston	0.70	1.9488	+1.9488	+0.67	32.1%	3.2638	5
Merton	0.73	2.0469	+2.0469	+0.70	9.5%	2.4631	6
GARCH-Merton	0.67	1.8507	+1.8507	+0.64	17.9%	2.4707	2
Heston-Merton	0.70	1.9223	+1.9223	+0.66	30.8%	3.1922	4

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE and Bias \$ are in stock-price dollars. ICP 25-75 is the share of historical S_t inside [p25, p75]. ABW = $\text{mean}(p75 - p25)$. Ranking uses RMSE% only.

Page 11 · Table block · MSFT · six models × four expiry windows

P-measure rolling cloud, reported path = p50. Bold green row = lowest RMSE% vs realized S_t. 1-day lookback, hourly rolling, Δt = 1 min, n_paths=10000, seed=42.

Table MSFT-9 MSFT · Jun 2023 listed expiry 2023-06-16
2023-06-09 09:30 → 2023-06-09 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.53	1.1695	-1.0324	-0.34	71.0%	3.0053	3
GARCH	0.57	1.3184	-1.2166	-0.40	70.8%	3.1625	4
Heston	0.52	1.1344	-0.9654	-0.32	77.9%	5.2426	2
Merton	0.92	2.6405	-2.6405	-0.87	12.3%	2.9426	6
GARCH-Merton	0.58	1.3466	-1.2625	-0.42	70.8%	3.4965	5
Heston-Merton	0.51	1.1015	-0.8452	-0.28	79.7%	5.8181	1

Table MSFT-10 MSFT · Jul 2023 listed expiry 2023-07-21
2023-07-14 09:30 → 2023-07-14 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.74	1.9539	+1.5201	+0.49	32.8%	3.3105	2
GARCH	1.00	2.5444	+2.3083	+0.74	27.9%	2.7403	5
Heston	0.75	1.9656	+1.5383	+0.49	75.4%	5.0424	3
Merton	0.85	2.1710	+1.8752	+0.60	29.7%	3.1900	4
GARCH-Merton	1.01	2.5591	+2.3465	+0.75	29.2%	3.3099	6
Heston-Merton	0.73	1.9332	+1.4987	+0.48	78.2%	5.0255	1

Table MSFT-11 MSFT · Aug 2023 listed expiry 2023-08-18
2023-08-11 09:30 → 2023-08-11 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.56	1.4949	-1.4944	-0.52	69.5%	3.4814	6
GARCH	0.54	1.3921	-1.3912	-0.48	87.4%	4.1206	4
Heston	0.53	1.4168	-1.4163	-0.49	85.4%	4.3415	1
Merton	0.54	1.4093	-1.4085	-0.49	71.0%	3.4264	3
GARCH-Merton	0.55	1.4196	-1.4188	-0.49	87.2%	4.0880	5
Heston-Merton	0.54	1.4308	-1.4302	-0.50	85.4%	4.3173	2

Table MSFT-12 MSFT · Sep 2023 listed expiry 2023-09-15
2023-09-08 09:30 → 2023-09-08 15:59 · n = 390 bars

Model	RMSE%	MAE	Bias \$	Bias%	ICP 25-75	ABW	Mark
GBM	0.91	2.3318	-2.3288	-0.81	28.2%	2.6502	4
GARCH	1.42	3.9742	-3.9742	-1.39	0.3%	1.8562	5
Heston	0.85	2.1254	-2.0823	-0.73	35.4%	3.7389	1
Merton	0.91	2.2799	-2.2565	-0.79	28.2%	2.5820	3
GARCH-Merton	1.42	3.9868	-3.9868	-1.39	0.3%	2.1662	6
Heston-Merton	0.86	2.1609	-2.1419	-0.75	34.1%	3.7541	2

RMSE% is $100 \times \sqrt{\text{mean}(((p50 - S_t)/S_t)^2)}$. MAE and Bias \$ are in stock-price dollars. ICP 25-75 is the share of historical S_t inside [p25, p75]. ABW = $\text{mean}(p75 - p25)$. Ranking uses RMSE% only.

Summary tables · RMSE% across companies and twelve windows

Table S1 · Overall ranking (mean RMSE% over 36 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 36 cells best
1	GBM	0.85	0.63	7
2	Heston-Merton	0.85	0.63	3
3	Heston	0.85	0.62	8
4	Merton	0.88	0.80	7
5	GARCH-Merton	0.92	0.66	6
6	GARCH	0.92	0.66	5

Table S2 · Wins by company (lowest RMSE% in that company×window)

Model	SPY wins	AAPL wins	MSFT wins	Total
GBM	3	2	2	7
GARCH	4	1	0	5
Heston	2	3	3	8
Merton	1	2	4	7
GARCH-Merton	2	3	1	6
Heston-Merton	0	1	2	3

Table S3 · Mean RMSE% by window (equal-weight mean of SPY / AAPL / MSFT)

Model	Oct22	Nov22	Dec22	Jan23	Feb23	Mar23	Apr23	May23	Jun23	Jul23	Aug23	Sep23	Mean of 12
GBM	3.34	0.48	0.52	0.50	0.96	0.77	0.91	0.70	0.46	0.42	0.60	0.55	0.85
GARCH	3.31	0.51	0.63	0.48	1.14	1.01	0.83	0.74	0.51	0.49	0.74	0.68	0.92
Heston	3.37	0.53	0.53	0.49	0.90	0.69	0.94	0.74	0.49	0.44	0.55	0.56	0.85
Merton	2.81	0.51	0.56	0.58	1.00	1.00	0.97	0.81	0.71	0.48	0.59	0.58	0.88
GARCH-Merton	3.28	0.50	0.62	0.48	1.14	1.02	0.84	0.74	0.51	0.49	0.74	0.68	0.92
Heston-Merton	3.36	0.52	0.53	0.49	0.92	0.72	0.93	0.73	0.49	0.44	0.57	0.56	0.85

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×window wins. A model can win the most cells without having the lowest average error (and vice versa).

Summary by company · mean RMSE% across twelve windows

Table S4.1 · SPY

Model	Mean RMSE%	Median # windows best	
GBM	0.67	0.46	3
GARCH	0.69	0.55	4
Heston	0.68	0.49	2
Merton	0.74	0.52	1
GARCH-Merton	0.69	0.54	2
Heston-Merton	0.68	0.49	0

Table S4.2 · AAPL

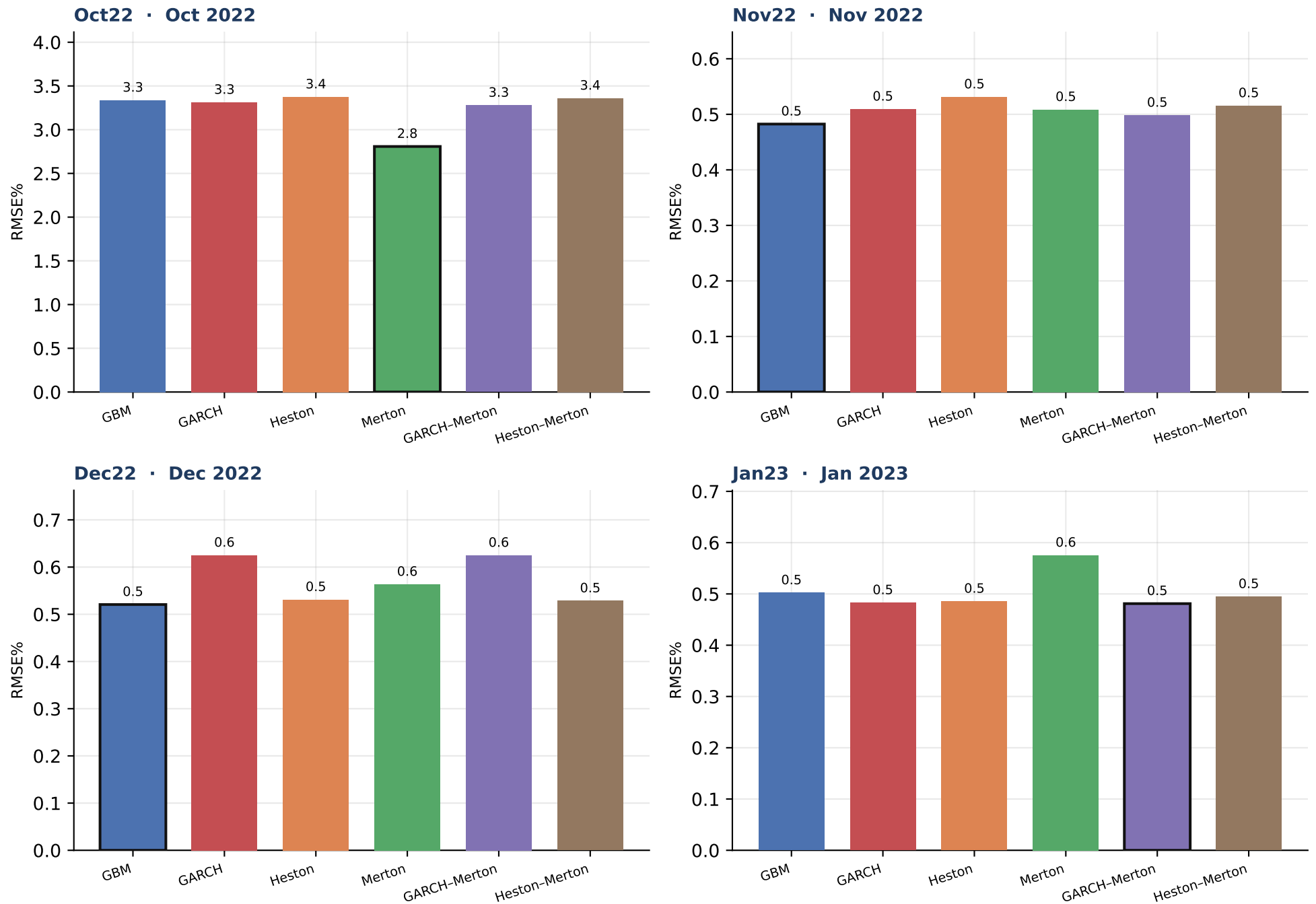
Model	Mean RMSE%	Median # windows best	
GBM	0.97	0.71	2
GARCH	1.08	0.86	1
Heston	0.97	0.68	3
Merton	0.98	0.87	2
GARCH-Merton	1.08	0.85	3
Heston-Merton	0.98	0.69	1

Table S4.3 · MSFT

Model	Mean RMSE%	Median # windows best	
GBM	0.91	0.68	2
GARCH	0.99	0.73	0
Heston	0.91	0.73	3
Merton	0.93	0.88	4
GARCH-Merton	0.99	0.73	1
Heston-Merto	0.91	0.71	2

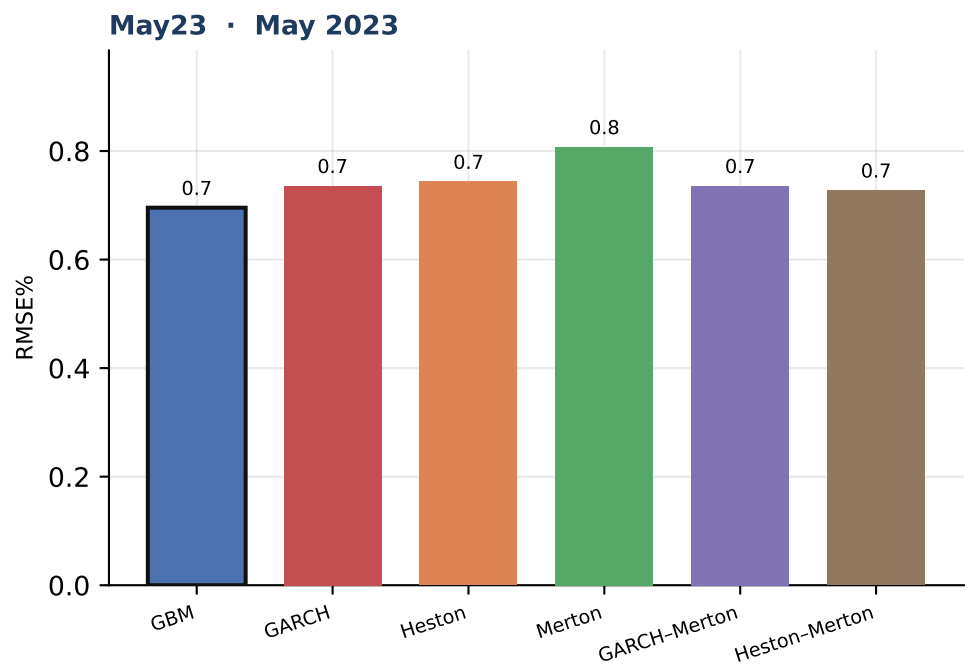
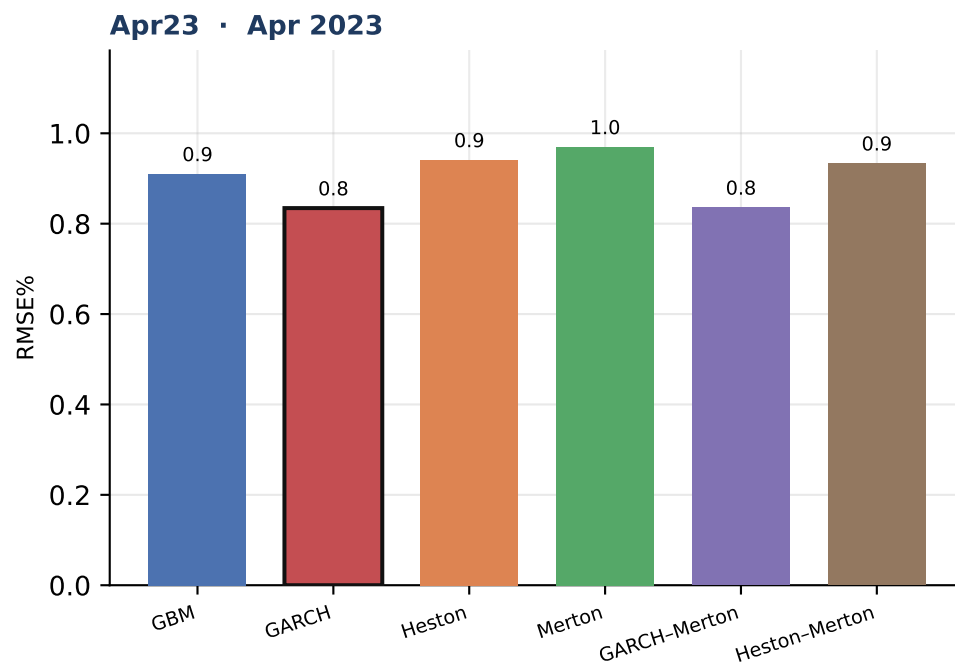
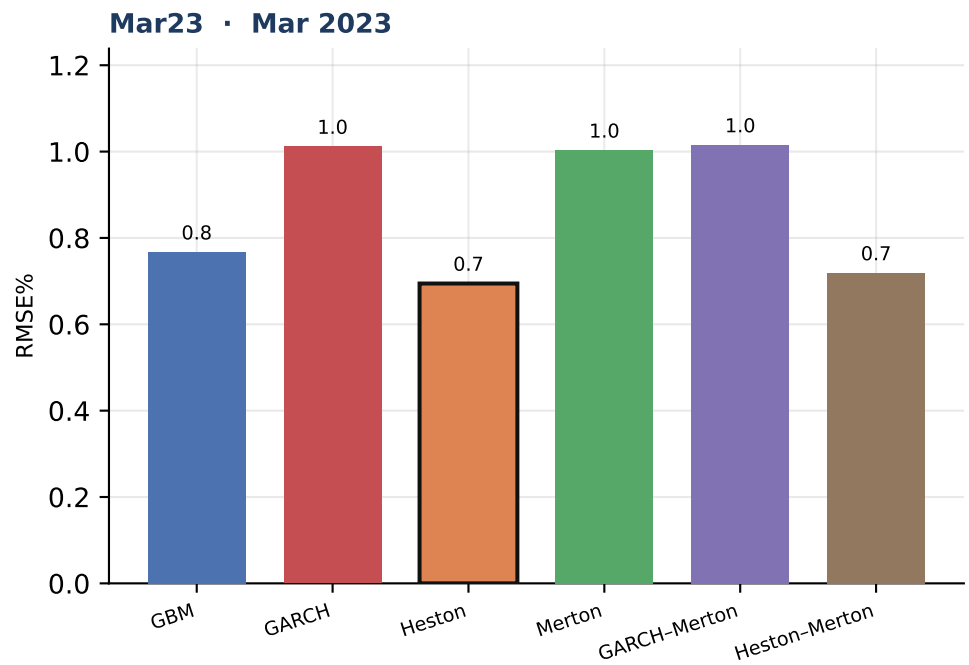
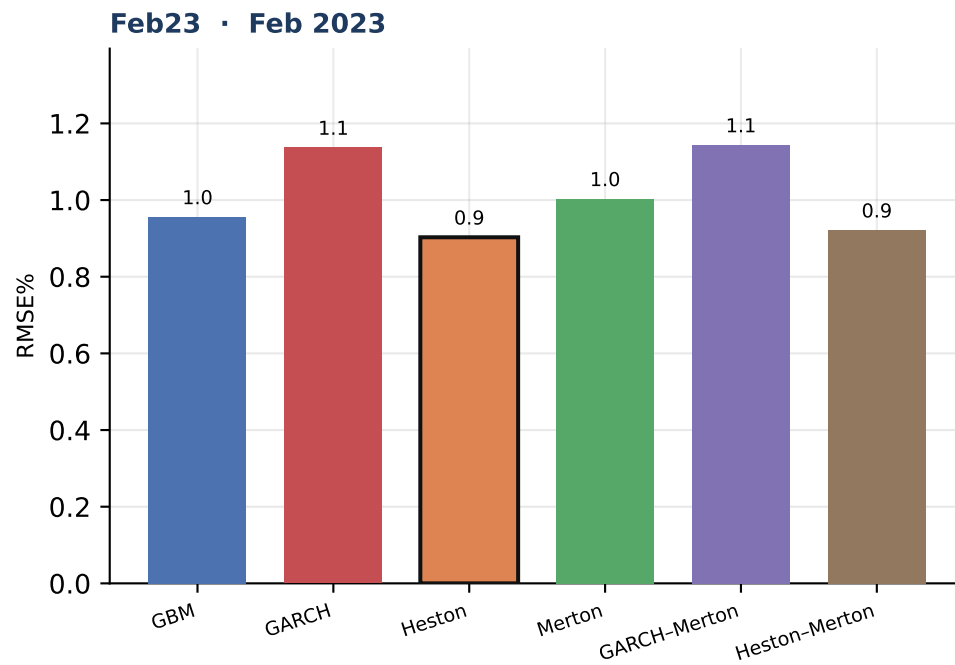
Each column is one company. Green row = lowest mean RMSE% across the twelve Friday-before-expiry windows for that name.

Figure 1 · RMSE% by model and window · mean of SPY / AAPL / MSFT



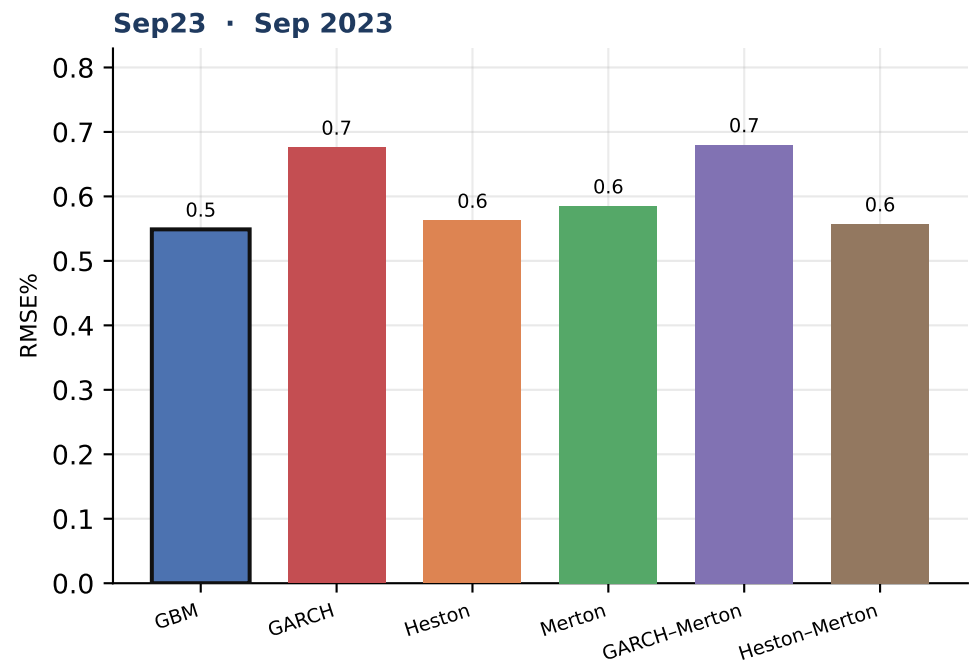
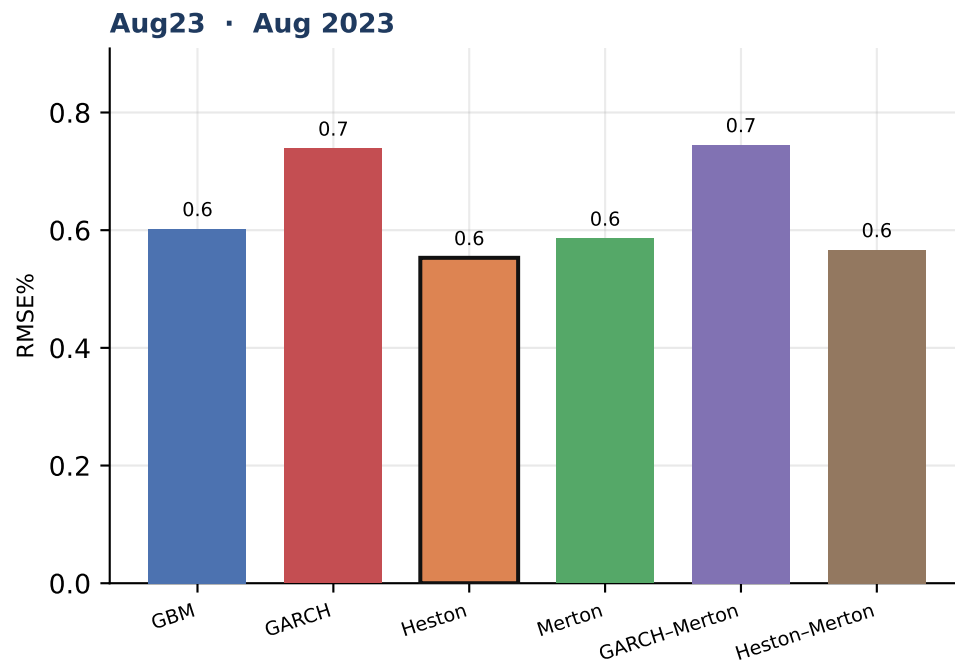
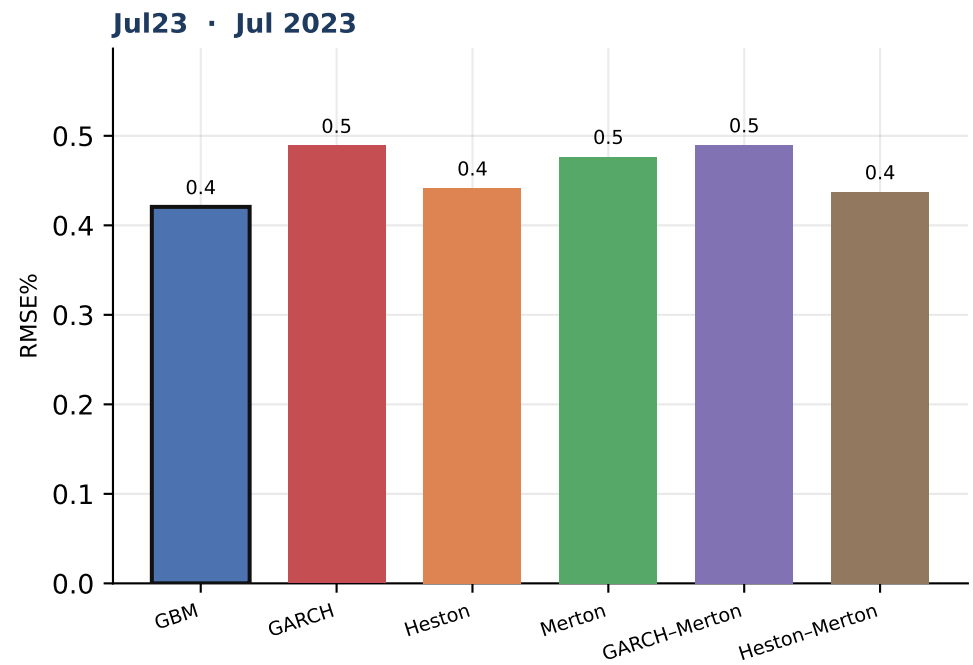
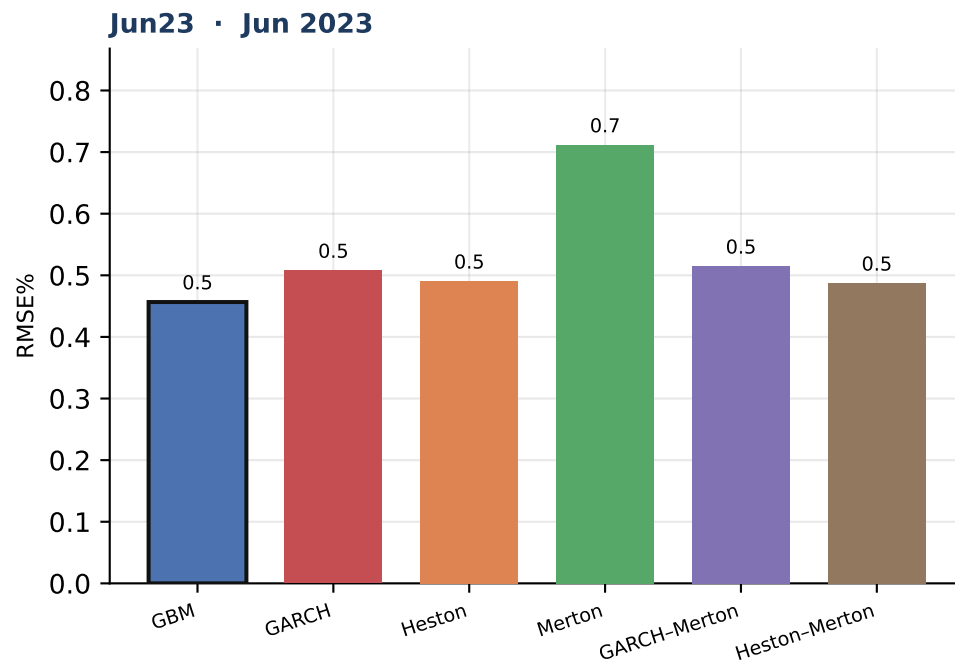
Outlined bar = lowest mean RMSE% in that window. Equal-weight mean of the three companies.

Figure 1 · RMSE% by model and window · mean of SPY / AAPL / MSFT



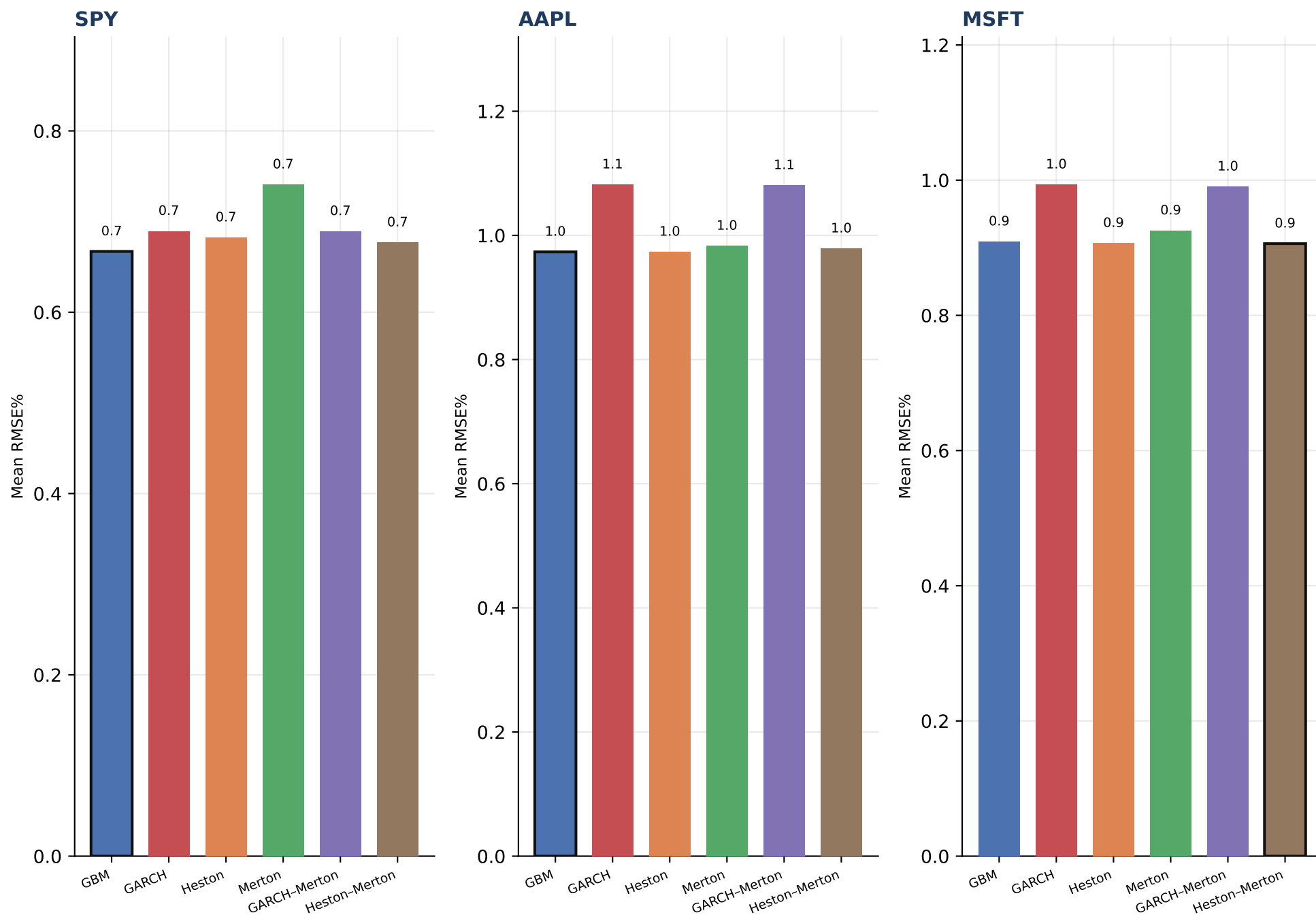
Outlined bar = lowest mean RMSE% in that window. Equal-weight mean of the three companies.

Figure 1 · RMSE% by model and window · mean of SPY / AAPL / MSFT



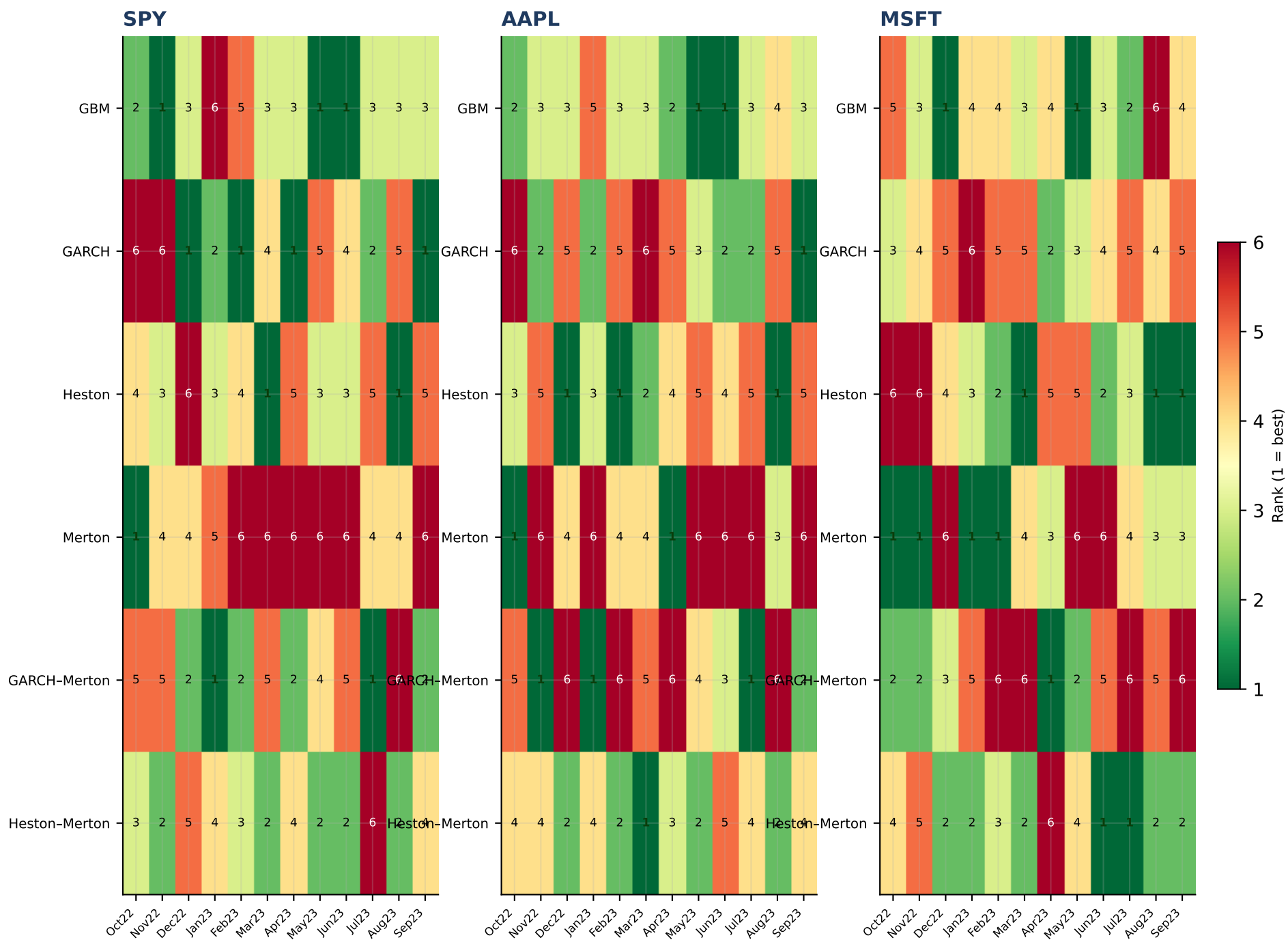
Outlined bar = lowest mean RMSE% in that window. Equal-weight mean of the three companies.

Figure 2 · RMSE% by model · each company (mean across twelve windows)



Outlined bar = lowest mean RMSE% for that company.

Figure 3 · Model rank by company × window (1 = best RMSE%)



Green / 1 = lowest RMSE% in that company×window cell. Rankings can and do change across names and windows.

Final analysis

Which models track realized S_t most closely?

- On mean RMSE% of the p50 path versus realized S_t across 36 company×window cells, GBM is best (mean RMSE% = 0.85; 7 of 36 cells).
- The ranking is not stable. Best-in-cell models: GARCH, GARCH-Merton, GBM, Heston, Heston-Merton, Merton.
- These conclusions are conditional on 1-day hourly P-measure calibration, 10000 paths, 1-minute steps to Friday 15:59, and the p50 scoring convention.